Supporting Information

A testing-effort artefact in the most-sensitive aquatic endpoint: structure-based PFAS screening recovers predictive validity but still cannot support clearance

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S1 Extraction from the ECOTOX Knowledgebase

All toxicity data derive from the U.S. EPA ECOTOX Knowledgebase ASCII release `ecotox_ascii_09_15_2026`, obtained from `https://gaftp.epa.gov/ecotox/`. The release contains 725,636 test records and 1,250,612 result records. We used it directly rather than through a curated derivative, so that the number of records per substance — the quantity at issue in this paper — reflects the primary compilation.

Selection proceeded in the order shown in Table S1. Tests were restricted to aquatic organisms in freshwater media; results were restricted to acute LC50, EC50 and IC50 and chronic NOEC, LOEC, LOEL, NOEL and MATC endpoints reported in mass-concentration units; and values carrying an inequality qualifier were separated out. Records in molar units (`uM`, `nM`, `mM`, `nmol/L`, `umol/L`) and in dose or body-weight units (`ppm diet`, `mg/kg bdwt`) were not converted and were excluded; molar units alone account for 3,967 of the rejected PFAS records.

Table S1: Record selection for the PFAS set.

Step	Count
Tests on PFASSTRUCTV6 CAS numbers	11,719
after restricting to aquatic organisms in freshwater	3,816
Results joined to those tests	29,375
with an acute or chronic endpoint of interest	24,379
with convertible mass-concentration units and a numeric value	16,357
carrying an inequality qualifier (excluded)	366
Usable uncensored records	**15,991**
Unique substances represented	103

The endpoint composition of the PFAS set is strongly asymmetric (Table S2) and this matters for the argument of the paper. Chronic sublethal endpoints outnumber acute endpoints by roughly fifteen to one, and they also lie lower on the concentration scale. A protective minimum taken over all strata is therefore drawn almost always from the chronic sublethal pool, and the size of that pool is what varies most between well- and poorly-studied substances.

The same filters applied without the PFAS restriction, and limited to acute endpoints, gave 106,095 freshwater acute records for 4,757 chemicals, which is the set used for the cross-chemical replication in Section S2 and as the training pool in Section S5.

Table S2: Endpoint composition of the 15,991 usable PFAS records.

Endpoint	Records	Endpoint	Records
NOEC	8,584	LC50	437
LOEC	6,360	NOEL	43
EC50	523	IC50	18
MATC	17	LOEL	9

S2 Endpoint constructions and the effort diagnostic

Four summaries of the same records were compared, all expressed as $y = \log_{10}(T/\mathrm{mg\,L^{-1}})$: the minimum over all strata (the protective convention), the median over the same records, the median over acute records only, and a depth-equalised minimum described in Section S3. Testing effort was taken as $\log_{10}$ of the number of usable records per substance.

Table S3: Association between each endpoint and testing effort. No model is involved; this is a property of the response variable.

Set	Endpoint	n substances	Spearman ρ vs $\log_{10}$ records
PFAS	minimum (all strata)	103	−0.734
PFAS	median (all strata)	103	−0.442
PFAS	acute median	70	−0.036
All chemicals	acute minimum	4,757	−0.462
All chemicals	acute median	4,757	−0.160

The last two rows are the cross-chemical replication. Restricting to acute data does not by itself remove the confound: within acute records alone, the minimum still carries roughly three times the association of the median. The confound attaches to the act of taking a minimum, not to the inclusion of chronic data, although the two compound in the PFAS set because the chronic pool is both larger and lower.

Table S4: Protective minimum by testing depth, PFAS set.

Records per substance	n	Median minimum	Median spread	Median of medians
1–3	26	+1.40	0.00	+1.46
4–10	15	−0.23	1.90	+0.72
11–50	26	−1.60	2.00	+0.24
51–300	26	−2.73	5.09	−0.82
>300	10	−4.96	7.22	−1.16

Spread is the within-substance range in $\log_{10}$ units and rises with depth as expected for a sample range. The final column shows that the median is far less sensitive to depth than the minimum, which is the basis for preferring it.

S3 Equal-depth subsampling

To separate the extreme-order-statistic artefact from genuine reverse causation, every substance with at least k records was subsampled to exactly k records without replacement, the minimum recomputed, and the correlation with the original record count recalculated. The procedure was repeated 200 times per k.

Table S5: Correlation between the protective minimum and testing effort after equalising sampling depth. Ranges are 2.5th–97.5th percentiles over 200 draws.

k	Eligible substances	Mean ρ	95% range
all records	103	-0.734	—
3	78	-0.311	$[-0.412, -0.207]$
5	70	-0.279	$[-0.387, -0.163]$
10	64	-0.293	$[-0.373, -0.204]$
20	47	-0.524	$[-0.600, -0.435]$

At $k = 3$, 5 and 10 the correlation stabilises near -0.29, slightly under half the value obtained using all records. The $k = 20$ row is higher, but its 47 eligible substances are by construction the well-studied ones, so the comparison is made over a restricted and shifted range; we report it for completeness and do not treat it as evidence against the pattern.

S4 Fixed-model comparison across endpoint constructions

In this experiment the broad non-PFAS model of the published analysis was held fixed and only the external target was changed, so that differences cannot be attributed to training, tuning or representation.

Table S6: External performance of one fixed model against four constructions of the same underlying records. Confidence intervals are substance-level bootstrap (4,000 resamples) on Spearman ρ.

Endpoint	n	R^2	ρ	95% CI on ρ	ρ vs effort
minimum (published)	103	-0.745	-0.015	$[-0.21, +0.18]$	-0.734
median	103	-0.147	$+0.076$	$[-0.13, +0.27]$	-0.442
depth-equalised minimum	70	-2.053	$+0.208$	$[-0.03, +0.41]$	-0.332
acute median	70	$+0.216$	$+0.424$	$[+0.21, +0.61]$	-0.036

The ordering of external performance follows the ordering of freedom from the effort confound. The depth-equalised minimum is an informative intermediate: its rank correlation improves over the raw minimum, consistent with removing part of the confound, but its R^2 is very poor because averaging subsampled minima produces a target with compressed variance that the model is not calibrated to. We report it as a diagnostic rather than as a candidate endpoint.

S5 Consistent rebuild

Training and external sets were rebuilt on the acute median. Model settings were those of the published analysis and were not re-tuned: 2,048-bit ECFP4 fingerprints (Morgan radius 2), random

forest with 300 trees, `max_features` $= \sqrt{p}$, seed 20260620. Broad cross-validation used five repeats of shuffled five-fold splitting with 160 trees per fold. Conformal validation used 40 repeats of the normalised split-conformal procedure given in the main text, with marginal calibration on a random 80/20 split of the broad set and PFAS-conditional calibration on disjoint halves of the PFAS set.

Table S7: Published protective endpoint versus the consistent acute-median rebuild. Model, representation and conformal procedure are identical.

Quantity	Protective minimum	Acute median
Broad training structures	3,788	3,718
External PFAS structures	119	58
Broad 5×5 CV R^2	0.387 ± 0.024	0.443 ± 0.029
External R^2	-0.580	$+0.367$
External Spearman ρ	-0.071	$+0.460$
95% CI	—	$[+0.22, +0.65]$
Marginal coverage at nominal 0.90	0.659	0.858
Marginal width	4.84	3.70
PFAS-conditional coverage	0.922	0.897
PFAS-conditional width	10.53	4.46
Inventory structures in domain	17,128	16,070

The external set is smaller on the acute median for two reasons: restricting to acute endpoints removes substances with only chronic data, and charge-neutralised desalting collapses salt forms of the same parent. Both reductions are deliberate, and the smaller external set is the principal limitation of the rebuilt analysis.

S6 Clearance across hazard thresholds

Because the two endpoints differ in protectiveness, a single threshold does not provide a comparable decision rule. On the 70 substances with both summaries the acute median lies $+2.69$ $\log_{10}$ units above the protective minimum, and 94.3% of measured PFAS exceed 0.1 mg/L on the acute median against 42.9% on the minimum. Clearance was therefore evaluated across a continuum.

Table S8: Clearance of in-domain inventory structures (n = 16,070) under the rebuilt model, by hazard threshold. The required half-width is the median over in-domain structures of the distance from the point prediction to the threshold; the achieved median half-width is 2.44.

Threshold	Cleared	%	Median required half-width
0.01 mg/L	14,991	93.3	$+3.22$
0.1 mg/L (used with the protective endpoint)	5,441	33.9	$+2.22$
1 mg/L (GHS acute aquatic Category 1)	306	1.9	$+1.22$
10 mg/L	0	0.0	$+0.22$
49 mg/L (matched to the protective endpoint)	0	0.0	-0.47
100 mg/L	0	0.0	-0.78

Two regimes are visible. Above 10 mg/L the required half-width approaches and then falls below zero, meaning the median in-domain structure is predicted more hazardous than the threshold and no narrowing of the interval would clear it. Below that, the barrier is interval width alone: at the

GHS Category 1 boundary the median structure would clear with a 90% interval of 2.43 $\log_{10}$ units against the 4.87 achieved.

S7 Structure standardisation and resolution

ECOTOX identifies substances by CAS number and carries no structures. CAS numbers were converted to the dashed form and resolved to SMILES through the PubChem PUG REST interface at the published rate limit of five requests per second; 4,311 of 4,757 chemicals resolved (90.6%), and the remainder were dropped. Resolution failures are not random with respect to chemical class — mixtures, poorly defined substances and some organometallics resolve less often — and we were not able to characterise the resulting bias.

Two standardisation choices differ from common practice and are stated explicitly because they change the data.

Desalting keeps the fluorine-bearing fragment, not the largest one. The usual rule retains the largest fragment by heavy-atom count. For tetrabutylammonium perfluoro-1-butanesulfonate the tetrabutylammonium cation (17 heavy atoms) exceeds the perfluorobutanesulfonate anion (13 heavy atoms), so the largest-fragment rule discards the analyte and models the counterion. We therefore retain the fluorinated fragment where one exists, falling back to the largest fragment otherwise.

Fragments are charge-neutralised before canonicalisation. Without this step a free acid and its salts produce different canonical SMILES and are counted as distinct structures. In the measured PFAS set, desalting alone collapses 138 rows to 131 parents, whereas desalting with neutralisation collapses them to 119. The difference is not cosmetic: perfluorooctanesulfonate appears in seven rows whose assigned protective values span 7.93 $\log_{10}$ units, and without neutralisation six of those rows survive as separate substances and can fall on both sides of a calibration split.

S8 OECD subclass assignment and statistical power

Substances were assigned to structure categories following the OECD (2021) definition and the 2018 OECD/UNEP structure series. The definitional gate requires at least one fully fluorinated, saturated methyl or methylene carbon carrying no hydrogen, chlorine, bromine or iodine. Carbons that are fully halogenated but not fully fluorinated (CHF_2, $CClF_2$), methine carbons (>CF–) alone, and unsaturated carbons (=CF_2) do not satisfy it.

Applying the gate to the measured set excluded seven substances, listed here because several would be retained by a looser fluorine-count rule: 4-bromo-3-chloro-3,4,4-trifluorobut-1-ene and 1,1,2-trichloro-1,2,2-trifluoroethane (halogens other than fluorine on the candidate carbon); perfluorobiphenyl, 2,3,4,5,6-pentafluoroaniline, decafluorobenzophenone and pentafluorobenzaldehyde (fluorine attached only to aromatic rings); and difluoroacetic acid, whose CHF_2 carbon bears a hydrogen.

Categories were assigned by parsing the molecule as C_nF_{2n+1}–X–R with head-group identity taking precedence over incidental fluorination. Assignment uses connectivity from fluorinated carbons rather than requiring a literal CF_2 bonded to the head group, so that perfluoroalkyl ether acids (for example HFPO-DA, whose α-carbon is a methine) and perfluorinated cycloalkane sulfonic acids are classified by head group rather than falling through to a residual category.

Six of seven intervals include zero. The seventh is one of seven comparisons and rests on

Table S9: External rank correlation by OECD structure class on the protective endpoint, where the measured set is largest. Intervals are substance-level bootstrap.

Class	n	Spearman ρ	95% CI
All OECD-2021 PFAS	112	+0.002	$[-0.18, +0.20]$
200a perfluoroalkane sulfonic	11	−0.611	$[-0.92, +0.03]$
100 perfluoroalkyl carboxylic	15	−0.206	$[-0.74, +0.47]$
700 side-chain fluorinated aromatic	12	−0.007	$[-0.52, +0.61]$
400 fluorotelomer	23	+0.274	$[-0.20, +0.63]$
200b perfluoroalkane sulfonamide	11	+0.323	$[-0.41, +0.87]$
600 other perfluoroalkyl	17	+0.350	$[-0.25, +0.71]$
500 perfluoroalkyl ether acid	13	+0.695	$[+0.14, +0.93]$

13 substances; at that sample size 80% power detects only $|\rho| \geq 0.71$, so a point estimate of +0.695 is approximately the smallest value the design could have returned as significant, and is the configuration most exposed to selection among the seven.

On the acute-median set the position is worse. Of 68 OECD-classified substances, three classes reach ten members and the largest reaches fourteen.

Table S10: Detectable effect size by class on the acute-median set, at 80% power and $\alpha = 0.05$.

Class	n	Detectable $\lvert\rho\rvert$
100 perfluoroalkyl carboxylic	14	0.69
200a perfluoroalkane sulfonic	14	0.69
400 fluorotelomer	11	0.76
700 side-chain fluorinated aromatic	9	0.82
600 other perfluoroalkyl	7	0.89
200b perfluoroalkane sulfonamide	6	0.92

Table S11: Measured substances required per class to resolve differential transfer, at 80% power.

Target effect size	$\alpha = 0.05$	$\alpha = 0.05/7$ (Bonferroni)
$\rho = 0.3$	85	133
$\rho = 0.4$	47	73
$\rho = 0.5$	29	44
$\rho = 0.6$	19	29

Resolving class-level differences at $\rho = 0.3$ across seven classes would require roughly 600 measured substances uncorrected, or 930 with correction, against the 103 the public freshwater record currently supports. We report this as the measurement requirement implied by the question rather than as an argument that the question is unanswerable.

S9 Files supporting the numerical audit

- `extract.py` — PFAS record extraction and per-substance summaries.
- `effort_test.py` — effort correlations and equal-depth subsampling (Tables S3–S5).

- `build_broad.py` — acute-median candidate set across all 4,757 chemicals.
- `resolve_smiles.py` — CAS to SMILES resolution via PubChem, resume-safe.
- `endpoint_fix.py` — fixed-model comparison across endpoint constructions (Table S6).
- `final_pipeline.py` — consistent rebuild: cross-validation, external validation, conformal intervals, clearance (Tables S7–S8).
- `oecd_classify.py` — OECD (2021) definitional gate and category assignment (Tables S9–S11).
- `make_fig_effort.py`, `make_fig_final.py`, `make_forest.py` — Figures 1–3.
- `ecotox_pfas_measured.csv`, `ecotox_acute_smiles.csv` — per-substance summaries with record counts.
- `endpoint_fix_results.csv`, `effort_subsample.csv`, `final_metrics.json` — values in Tables S5–S8.